Tema 4

1.5

	CR	CPi
P1	3GHz	1,5
P2	2,5GHz	1,0
P3	4,0GHz	2,2

a) Highest performance

P1: $\frac{3 \cdot 10^9}{1,5} = 2 \cdot 10^9$ instr/s

P2: $\frac{2,5 \cdot 10^9}{1,0} = 2,5 \cdot 10^9$ instr/s - highest

P3: $\frac{4 \cdot 10^9}{2,2} = 1,81 \cdot 10^9$ instr/s

b) each processor execute for 10s
find the number of cycles and nr. of instr.

P1: $20 \cdot 10^9$ instr

$CC = CPi \cdot IC = 1.5 \cdot 20 \cdot 10^9 = 30 \cdot 10^9$

P2: $25 \cdot 10^9$ instr.

$CC = 1 \cdot 25 \cdot 10^9 = 25 \cdot 10^9$

P3: $18,18 \cdot 10^9$ instr

$CC = 2.2 \cdot 18,18 \cdot 10^9 = 40 \cdot 10^9$

c) time $\searrow$ 30% CPi $\nearrow$ 20%

CR = ?

$time = \frac{ic \cdot CPi}{CR}$

$\Rightarrow CR = \frac{ic \cdot CPi}{time}$

$new\ CR = \frac{ic \cdot (1.2)\ CPi}{(0,7)\ time} = 1,714$ clock rate